(1) Show that $(\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$

Solution : We know that $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

$$\text{L. H. S} = (\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3\cos^2\theta(i\sin\theta) + 3\cos\theta(i\sin\theta)^2 + (i\sin\theta)^3$$

$$= \left(\cos^3\theta - 3\cos\theta\sin^2\theta\right) + i\left(3\cos^2\theta\sin\theta - \sin^3\theta\right)$$

$$= \left(\cos^3\theta - 3\cos\theta(1-\cos^2\theta)\right) + i\left(3(1-\sin^2\theta)\sin\theta - \sin^3\theta\right)$$

$$= \left(\cos^3\theta - 3\cos\theta + 3\cos^3\theta\right) + i\left(3\sin\theta - 3\sin^3\theta - \sin^3\theta\right)$$

$$= \left(4\cos^3\theta - 3\cos\theta\right) + i\left(3\sin\theta - 4\sin^3\theta\right)$$

$$= \cos 3\theta + i\sin 3\theta$$

$$\therefore\ (\cos\theta + i\sin\theta)^3 = \cos 3\theta + i\sin 3\theta$$

(2) Show that $\cos 3\theta - \sin 3\theta = (\cos\theta + \sin\theta)(1 - 2\sin 2\theta)$

Solution : L.H.S. $= \cos 3\theta - \sin 3\theta$

$= 4\cos^3\theta - 3\cos\theta - (3\sin\theta - 4\sin^3\theta)$

$= 4\cos^3\theta - 3\cos\theta - 3\sin\theta + 4\sin^3\theta$

$= 4\cos^3\theta + 4\sin^3\theta - 3\cos\theta - 3\sin\theta$

$= 4(\cos^3\theta + \sin^3\theta) - 3(\cos\theta + \sin\theta)$

$= 4(\cos\theta + \sin\theta)(\cos^2\theta - \cos\theta\sin\theta + \sin^2\theta) - 3(\cos\theta + \sin\theta)$

$[\because a^3 + b^3 = (a+b)(a^2 - ab + b^2)]$

$= (\cos\theta + \sin\theta)[4(\cos^2\theta + \sin^2\theta - \sin\theta\cos\theta) - 3]$

$= (\cos\theta + \sin\theta)[4(1 - \sin\theta\cos\theta) - 3]\ (\because \cos^2\theta + \sin^2\theta = 1)$

$= (\cos\theta + \sin\theta)(4 - 4\sin\theta\cos\theta - 3)$

$= (\cos\theta + \sin\theta)(1 - 2 \times 2\sin\theta\cos\theta)$

$= (\cos\theta + \sin\theta)(1 - 2\sin 2\theta)$

$=$ RHS

$\therefore$ LHS $=$ RHS

$\therefore \cos 3\theta - \sin 3\theta = (\cos\theta + \sin\theta)(1 - 2\sin 2\theta)$

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(3) Prove that $\cos^2\theta+\cos^2(\theta+120^0)+\cos^2(\theta-120^0)=\frac{3}{2}$

Solution : We know that $\cos(A+270^0)=\sin A$ and $\cos(A-270^0)=-\sin A$

L.H.S. $=\cos^2\theta+\cos^2(\theta+120^0)+\cos^2(\theta-120^0)$

$=\frac{1}{2}\left[2\cos^2\theta+2\cos^2(\theta+120^0)+2\cos^2(\theta-120^0)\right]$

$=\frac{1}{2}\left[1+\cos(2\theta)+1+\cos(2\theta+240^0)+1+\cos(2\theta-240^0)\right]$

$=\frac{1}{2}\left[3+\cos(2\theta)+\cos(2\theta-30^0+270^0)+\cos(2\theta+30^0-270^0)\right]$

$=\frac{1}{2}\left[3+\cos(2\theta)+\sin(2\theta-30^0)-\sin(2\theta+30^0)\right]$

$=\frac{1}{2}\left[3+\cos(2\theta)+\sin(2\theta-30^0)-\sin(2\theta+30^0)\right]$

$=\frac{1}{2}\left[3+\cos^2\theta+2\cos\left(\frac{2\theta-30^0+2\theta+30^0}{2}\right)\sin\left(\frac{2\theta-30^0-2\theta-30^0}{2}\right)\right]$

$\left(\because \sin C-\sin D=2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)\right)$

$=\frac{1}{2}\left[3+\cos^2\theta+2\cos\left(\frac{4\theta}{2}\right)\sin\left(\frac{-60^0}{2}\right)\right]$

$=\frac{1}{2}\left[3+\cos^2\theta+2\cos^2\theta\sin(-30^0)\right]$

$=\frac{1}{2}\left[3+\cos^2\theta+2\cos^2\theta\times-\sin 30^0\right]$ $(\because \sin(-\theta)=-\sin\theta)$

$=\frac{1}{2}\left[3+\cos^2\theta-2\cos^2\theta\times\frac{1}{2}\right]$ $\left[\because \sin 30^0=\frac{1}{2}\right]$

$=\frac{1}{2}\left[3+\cos^2\theta-\cos^2\theta\right]$

$=\frac{3}{2}$

$=$ RHS.

(4) If $A+B+C=\pi$ then prove that $\cos^2A+\cos^2B+\cos^2C=1-2\cos A\cos B\cos C$.

Hence prove that $\cos 60^0=\frac{1}{2}$.

Solution : L.H.S. $=\cos^2A+\cos^2B+\cos^2C$

$=\frac{1}{2}\left(2\cos^2A+2\cos^2B+2\cos^2C\right)$

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$$=\frac{1}{2}\left(1+\cos 2A+1+\cos 2B+2\cos^2 C\right)$$

$$=\frac{1}{2}\left(2+\cos 2A+\cos 2B+2\cos^2 C\right)$$

$$=\frac{1}{2}\left(2+2\cos(A+B)\cos(A-B)+2\cos^2 C\right)$$

$$=\frac{1}{2}\left(2+2\cos(\pi-C)\cos(A-B)+2\cos^2 C\right)$$

$$=\frac{1}{2}\left(2-2\cos C\cos(A-B)+2\cos^2 C\right)$$

$$=\frac{1}{2}\left(2-2\cos C[\cos(A-B)-\cos C]\right)$$

$$=\frac{1}{2}\left(2-2\cos C[\cos(A-B)+\cos(A+B)]\right)$$

$$=\frac{1}{2}\left(2-4\cos C\cos A\cos B\right)$$

$$=1-2\cos A\cos B\cos C$$

= R.H.S.

In the above take $A=B=C=60^0$ and find the value of $\cos 60^0$.

Now: $\cos^2 A+\cos^2 B+\cos^2 C = 1-2\cos A\cos B\cos C$

Let $A=B=C=60^0$

$\therefore 3\cos^2 A = 1-2\cos^3 A$

Put $\cos A = x$

$\therefore 3x^2 = 1-2x^3$

$\therefore 2x^3+3x^2-1=0$

If $x=-1$, LHS $=2(-1)^3+3(-1)^2-1$
$=-2+3-1=0$

$\therefore (x+1)$ is one factor of LHS

$\therefore$ By synthetic division.

$\therefore$ Factors of LHS $=(x+1)(2x^2+x-1)=0$ — ①

But $2x^2+x-1 = 2x^2+2x-x-1$

From equation ① $=(x+1)(2x-1)$

$(x+1)^2(2x-1)=0$

$\therefore x=-1$ or $x=\frac{1}{2}$

but $x=\cos A$ where $A=60^0$

$\therefore \cos A=\cos 60^0=-1$ or $\cos A=\cos 60^0$

but $\cos 60^0 \neq -1 \Rightarrow \cos A=\cos 60^0=\frac{1}{2}$

(5) Find the values of $\sin 18^0$ and $\cos 18^0$.

Solution : Let $\theta = 18^0$

$5\theta = 5\times 18^0 = 90^0$

$5\theta = 90^0$

$2\theta+3\theta = 90^0$

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$\therefore 2\theta = 90^\circ - 3\theta$

$\therefore \sin 2\theta = \sin(90^\circ - 3\theta)$

$\therefore \sin 2\theta = \cos 3\theta$

$(\because \sin(90^\circ - \theta) = \cos\theta)$

$\therefore 2\sin\theta\cos\theta = 4\cos^3\theta - 3\cos\theta$

$\therefore 2\sin\theta\cos\theta - 4\cos^3\theta + 3\cos\theta = 0$

$\therefore \cos\theta(2\sin\theta - 4\cos^2\theta + 3) = 0$

$\therefore 2\sin\theta - 4\cos^2\theta + 3 = 0$

$\therefore 2\sin - 4(1 - \sin^2\theta) + 3 = 0$

$\therefore 2\sin\theta - 4 + 4\sin^2\theta + 3 = 0$

$\therefore 4\sin^2\theta + 2\sin\theta - 1 = 0$

Let $\sin\theta = x$

$4x^2 + 2x - 1 = 0$

by formula method

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-2 \pm \sqrt{4+16}}{8}$$

$$\therefore x = \frac{-2 \pm \sqrt{20}}{8}$$

$$\therefore x = \frac{-2 \pm 2\sqrt{5}}{8}$$

$$\therefore x = \frac{-1 \pm \sqrt{5}}{4}$$

$$\therefore \sin\theta = \sin 18^\circ = \frac{-1 \pm \sqrt{5}}{4}$$

Since sin is positive

$$\therefore \sin 18^\circ = \frac{-1 + \sqrt{5}}{4}$$

$(\sin u \sqrt{5} = 2.2361)$

we know that

$\sin^2 18^\circ + \cos^2 18^\circ = 1$

$\therefore \cos^2 18^\circ = 1 - \sin^2 18^\circ$

$\therefore \cos 18^\circ = \sqrt{1 - \sin^2 18^\circ}$

$$\therefore \cos 18^\circ = \sqrt{1 - \left(\frac{-1 + \sqrt{5}}{4}\right)^2}$$

$$\therefore \cos 18^\circ = \sqrt{1 - \left(\frac{1 - 2\sqrt{5} + 5}{16}\right)}$$

$$\cos 18^\circ = \sqrt{\frac{16 - 6 + 2\sqrt{5}}{16}}$$

$$\therefore \cos 18^\circ = \sqrt{\frac{10 + 2\sqrt{5}}{16}}$$

$$\therefore \boxed{\cos 18^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}}$$

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(6) If $\tan A = \frac{5}{6}$ and $\tan B = \frac{1}{11}$ then show that $A + B = \frac{\pi}{4}$.

Solution :

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A+B) = \frac{\frac{5}{6} + \frac{1}{11}}{1 - \frac{5}{6} \times \frac{1}{11}}$$

$$\therefore \tan(A+B) = \frac{\frac{55+6}{66}}{1 - \frac{5}{66}}$$

$$\therefore \tan(A+B) = \frac{\frac{61}{66}}{\frac{66-5}{66}}$$

$$\therefore \tan(A+B) = \frac{\frac{61}{66}}{\frac{61}{66}}$$

$$\therefore \tan(A+B) = 1$$

but $\tan\left(\frac{\pi}{4}\right) = 1$

$$\therefore \tan(A+B) = \tan\left(\frac{\pi}{4}\right)$$

$$\therefore A+B = \frac{\pi}{4}$$

Hence proved.

Sign of Teacher :

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